

NATS CLI Cheatsheet

Overview

Download: <https://github.com/nats-io/natscli>
One stop command line utility to interact with and manage NATS.
Independent from NATS server version.
Uses only public Golang API.
Please update regularly or compile the GitHub main branch.

Getting started

Built-in cheat sheet
nats cheat
Always check help for additional options
nats server info --help
Quick system status
nats server list
Show raw nats cli messages. No secret sauce
nats server list --trace

Publish/Subscribe

Listen to a subject
nats sub foo
Publish to a subject
nats pub foo Hello
Listen to wildcards
nats sub '>'
nats sub 'orders.emea.*'
nats sub 'orders.*.o1234'

Context and credentials

List local connection contexts
nats context list
Store credentials and server URL for *cloud*
nats context add --server
nats://mycloud.io --user
admin --password admin cloud
Select context
nats context select cloud

Request/Reply

Simple reply mockup
nats reply bar World
Send a request
nats request bar Hello

JetStream

List streams, including offline and inaccessible streams
nats stream list
Create stream interactively and with defaults
nats stream add <stream>
nats stream add <stream>
--replicas 3 --defaults
Stream view content
nats stream view <stream>
Publish JetStream message with ack
nats pub --jetstream foo Hello

Key-Value Store

List Key-Value buckets
nats kv list
Create a KV bucket
nats kv add <bucket>
Put and get a key
nats kv put <bucket> foo Hello
nats kv get <bucket> foo
Watch for changes
nats kv watch <bucket> '>'

Distributed security

Create an operator
nats auth operator add
my-system
Generate a sample config
nats server generate
server.conf
Walkthrough with accounts and users
nats auth cheat

Server Health and Monitoring

Show system health (SYS)
nats server list
Detailed server information (SYS)
nats server info <server>
Check account traffic
nats traffic
Gather and analyze for debugging (SYS)
nats audit gather
nats audit analyze <audit zip>

JetStream Monitoring

Show JetStream and meta status (SYS)
nats server report jetstream
--watch=1
List all streams
nats stream report
Stream details
nats stream info <stream>

JetStream backup and purge

Backup a stream with consumer state
nats stream backup <stream> <dir>
Restore stream from backup
nats stream restore <file>
Backup just messages
nats stream backup --no-consumers
<stream> <dir>
Backup all streams in current account
nats account backup <dir>
Purge a stream
nats stream purge <stream>
Purge a subject in a stream
nats stream purge --subject foo
<stream>

Trace and debug message flow

Trace message without delivering to clients
nats trace foo Hello
Trace with delivery and timestamps
nats trace --timestamp --deliver foo

Measure roundtrip and latency

Roundtrip single client to server
nats --server srv1:4222 rtt
Measure latency between clients connected to different servers
nats latency --server srv1:4222
--server-b srv2:4222

Benchmarking

Subscribe core NATS for 5 clients
nats bench sub foo --clients 5
--msgs 10000
Publish core NATS with 10 publishers
nats bench pub foo --clients 10
--msgs 10000 --size 512
Request-reply server for core nats
nats bench service serve --clients 5
testservice
Benchmark core nats subscribe for 5 clients
nats bench service request --clients
5 testservice --msgs 10000
JetStream async acknowledged publishing of batches of 100
nats bench js pub foo --create
--batch 100
JetStream sync publishing using 10 clients - purge stream
nats bench js pub foo --purge
--batch=1 --clients=10 --msgs 10000
JetStream delivery from a stream through a durable consumer with 4 instances
nats bench js consume --clients 4
--msgs 10000

Dashboards - top style

Watch all servers (SYS)
nats server watch servers
Watch a single server with statistics (SYS)
nats top <server-name>